



Potential Questions Document

WRITTEN DELIVERABLE





Regen Stride

RESILIENCE, WOVEN INTO EVERY STEP

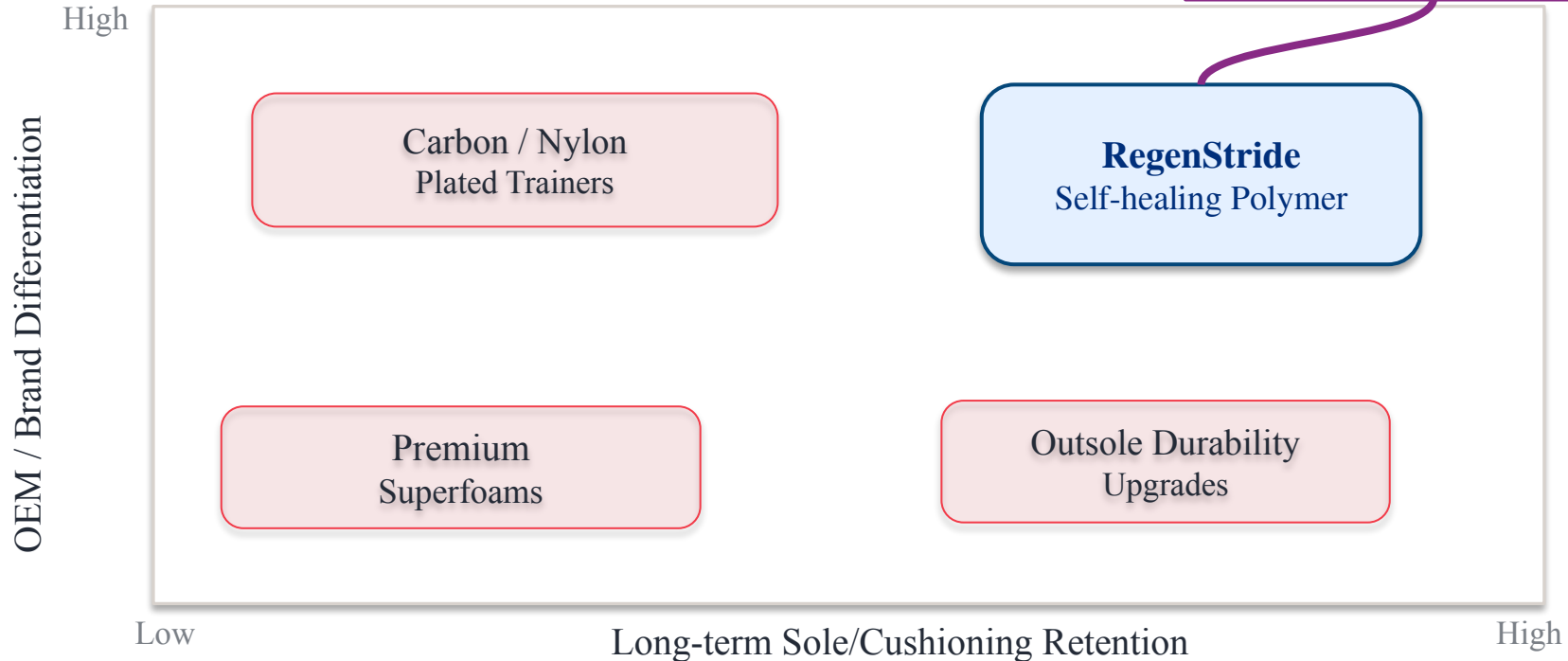
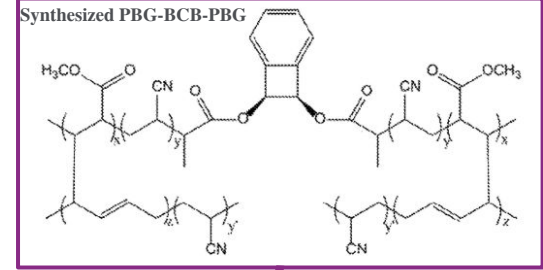
A self-healing polymer that extends the lifespan of athletic footwear.

TEAM 3
POW-WOW-Polymers

After about 400 miles, runners discard their sneakers because the midsole loses its cushioning and protective performance, long before the rest of the shoe wears out.



Technical Competitive Advantage



Business Model

License polymer formulation to footwear OEMs. Earn royalties as unit volume scales.

Licensed Technology
Self-healing polymer

The Customer
Athletic footwear OEMs

Revenue Model
8-10% Royalty
per OEM unit sold

1

Validate
2x Lifecycle
Target



2

Partner
License
Agreement



3

Launch
Limited Halo
Drop



4

Monetize
OEM volume →
royalties scale

Value Proposition

Nike OEM Design Engineering Team

Only material on the market offering retained cushioning as a licensed IP

- Ad. point: certified impact protection beyond 400 miles
- Premium brand positioning
- Fewer returns



End Consumer: Runner

- Fewer injuries
- Social proof from celebrities
- Fewer trips to shoe store



The Team



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GTM Strategy

Why Nike first?

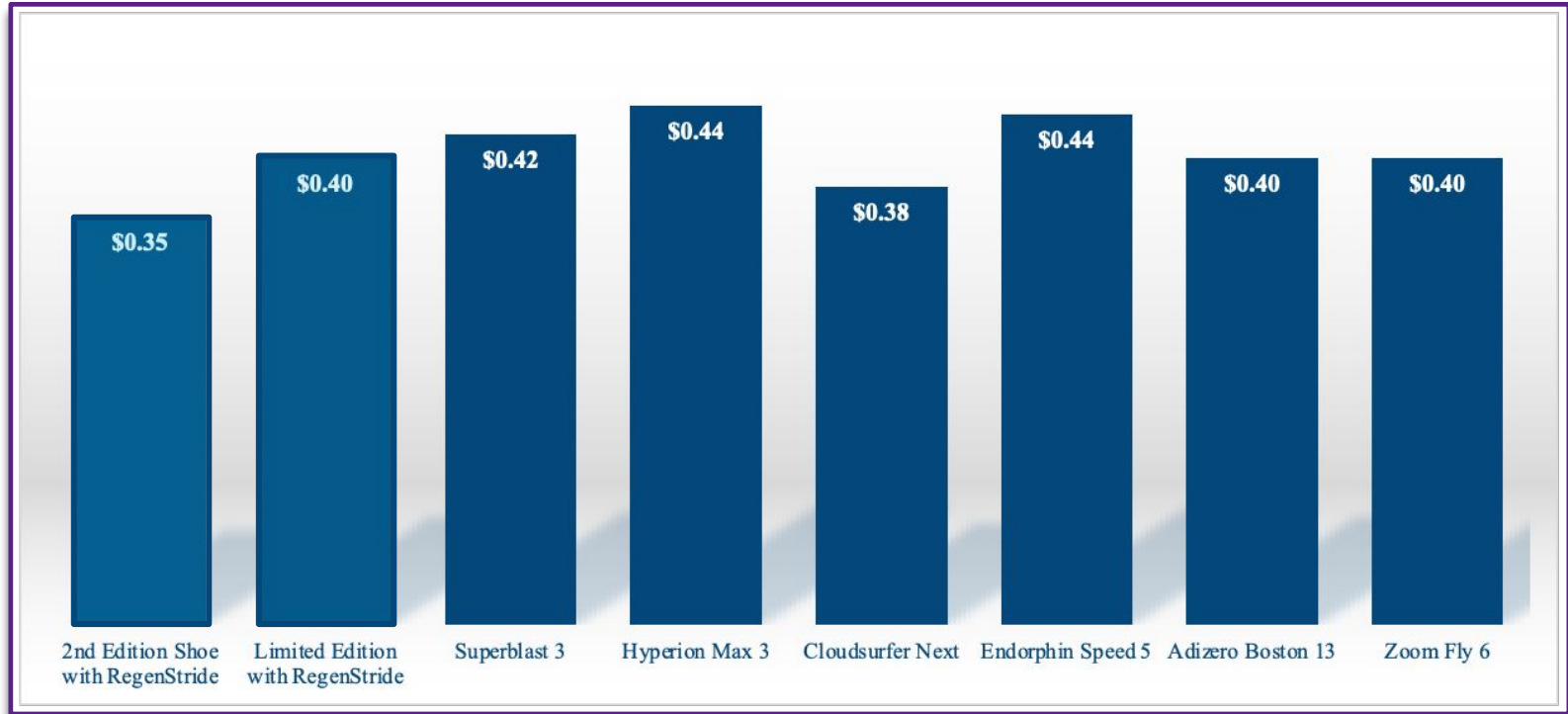
- Deep R&D infrastructure
- Established limited edition playbook
- Global branch reach to amplify Phase 2 mass market adoption



Competitive Analysis: Competitor Value Matrix

Technology	Products	Retained Cushioning	OEM Differentiation	Lifetime Value Story
Premium Superfoams	ASICS Superblast; Nike ZoomX; adidas Lightstrike	Medium	Low	Low
Carbon / Nylon Plated Trainers	Zoom Fly 6; Endorphin Speed 5; Hyperion Max 3	Low	High	Medium
Outsole Durability Upgrades	Boston 13 and other high-rubber premium	Low	Low	Medium
RegenStride Self-Healing Polymer	RegenStride Technology OEM Partner	High	High	Medium

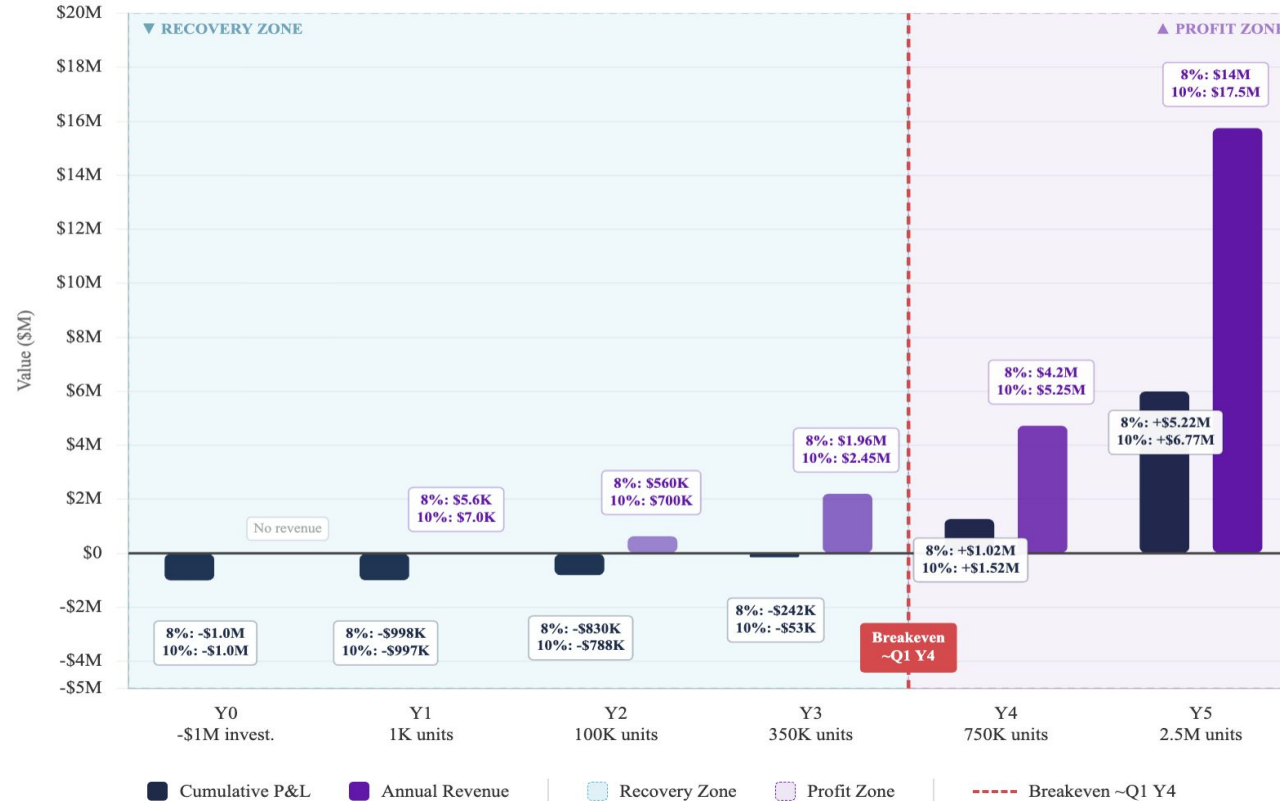
Competitive Analysis: Cost per Mile Comparison



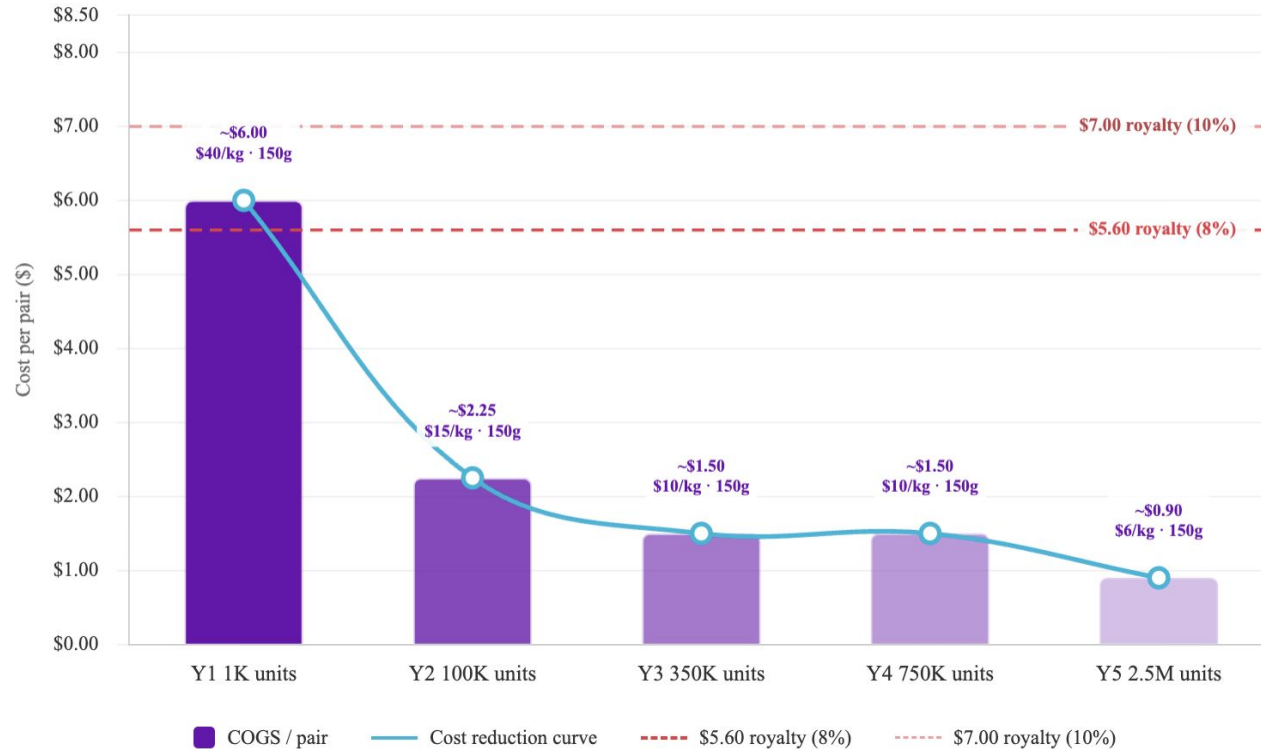
Milestones



Financial Projections: Revenue & Profit



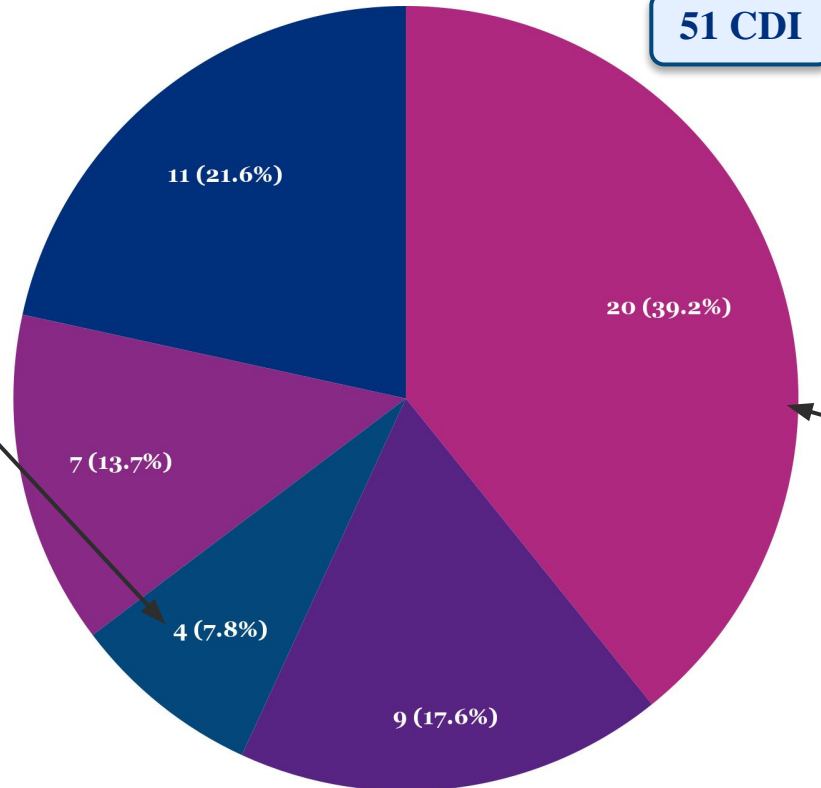
Material Cost Reduction Over Time



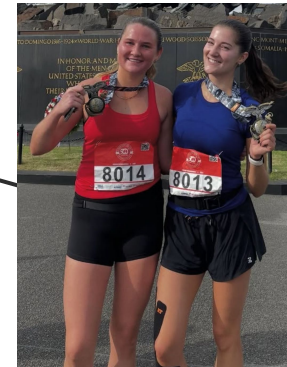
Customer Discovery Breakdown



Aaron Hankins,
Senior Systems and
Process Development,
Nike

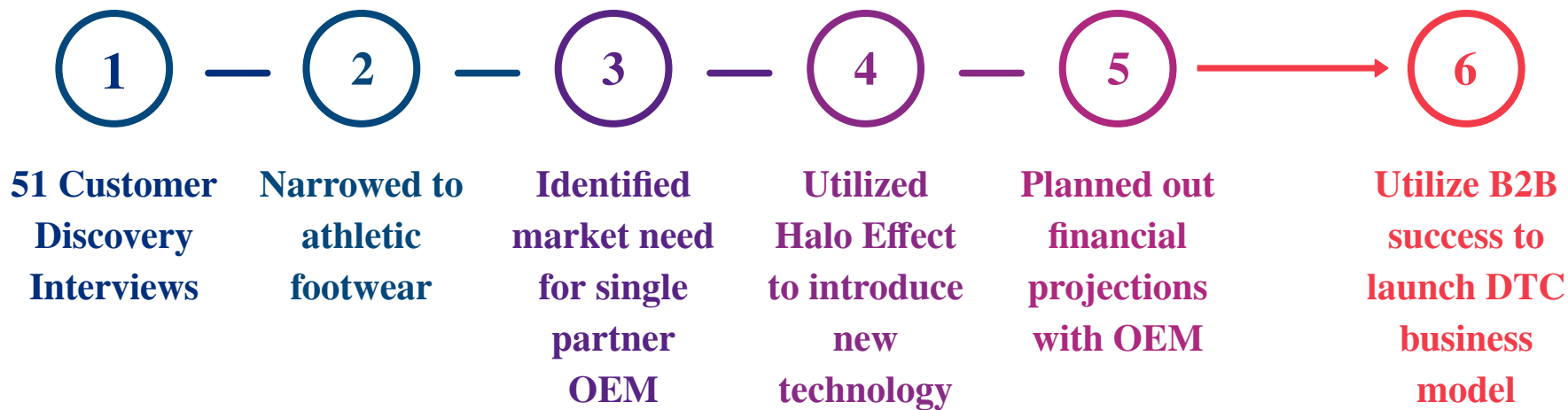


- Runners & Athletes
- Materials & Polymer Experts
- Footwear Product Developers
- Manufacturing & Engineers
- Commercial & Strategy Professionals



**Elizabeth Fine & Caroline
Lamb, Cornell Students and
3x marathon runners**

What We've Accomplished and Future Plans



Established a business model →
CAC 1:7.5 LTV

Funding Roadmap

Pre-Seed (Now)
Months 0-18

Seed
Months 18-36

Series-A
Months 36-60

\$750K-\$1M

Polymer optimization · OEM pilot ·
Third-party validation

Funding Sources:

- NSF SBIR/STTR Phase I
- Cornell Tech / eLab network
- Material Impact Ventures
- SOSV/HAX
- Athletic footwear strategic partners
- Cornell alumni angels

Gate: Validated footwear prototype + OEM
pilot agreement

\$2M-\$5M

Commercial Integration · Manufacturing
scale-up · Licensing execution

Funding Sources:

- Footwear/consumer strategic investors
- Climate-tech & advanced materials
VCs
- Nike/Adidas innovation partnerships
- NY Ventures / Empire State
Development
- Eclipse Ventures

Gate: First commercial footwear launch +
recurring royalty revenue

\$8M-\$15M

Multi-SKU scaling · Additional OEM
expansion · Scaled licensing deployment

Funding Sources:

- Strategic CVS (Nike, Adidas,
Puma innovation arms)
- Advanced materials growth funds
- Consumer product growth equity
- Institutional venture firms

Gate: Scaled recurring licensing revenue
across footwear categories

Thanks!
Questions?

Appendix I: Business Model Canvas

As validated from our 51 interviews

Key Partners

- Exclusive OEM partner (integration & revenue share)
- Licensing / manufacturing partner (royalties)
- Validation labs (spec-in)

Key Activities

- Polymer formulation & optimization
- Impact/fatigue testing under cyclic loading
- Integration into midsole materials
- Prototype footwear testing

Value Proposition

- Reduces midsole material and structural material fatigue
- Maintains cushioning and energy return under loading
- Extends usable lifespan of performance footwear
- Enables premium long-lasting running shoes

Customer

Relationships

- Direct OEM engineer engagement & evaluation
- Support through sampling, testing, validation
- Spec-in → expand & long-term agreements

Channels

- Direct virtual outreach to OEM design engineers
- Prototype validation
- Direct commercial path to OEMs
- Licensing pathway through strategic partner

Key Resources

- Self-healing polymer IP
- Polymer formulation for elastomers/foam
- Cyclic fatigue & compression testing

Cost Structure

- R&D & testing
- Raw polymer inputs
- Manufacturing scale-up
- Prototype footwear integration
- IP protection

Revenue Streams

- B2B Licensing (% of OEM sales)
- Pilot / prototype fees
- DTC Production revenue share

Customer Segments

- Athletic footwear materials engineers
- Footwear product development teams at athletic brands
- Innovation/advanced materials teams at sporting shoe brands

Appendix II: Competitive Analysis Calculations

Shoe	Brand	Cost	Material / Tech	Useful Life	Cost/Mile
2nd Edition Shoe with RegenStride	Exclusive OEM Partner	\$350	RegenStride proprietary	1000	\$0.35
Limited Edition with RegenStride	Exclusive OEM Partner	\$900	RegenStride proprietary	2250	\$0.40
Superblast 3	ASICS	\$210	FF LEAP cushioning	500	\$0.42
Hyperion Max 3	Brooks	\$200	DNA FLASH v2 + PEBA DNA GOLD + nylon SpeedVault plate	450	\$0.44
Cloudsurfer Next	On	\$150	CloudTec Phase + Helion foam	400	\$0.38
Endorphin Speed 5	Saucony	\$175	PWRRUN PB + semi-rigid nylon plate	400	\$0.44
Adizero Boston 13	adidas	\$160	Lightstrike/Lightstrike Pro + EnergyRods 2.0	400	\$0.40
Zoom Fly 6	Nike	\$180	ZoomX + full-length carbon fiber Flyplate	450	\$0.40

- [1] ASICS, “SUPERBLAST 3,” ASICS United States.
- [2] RunRepeat, “ASICS Superblast 3 Review (2026).”
- [3] Brooks Running, “Hyperion Max 3.”
- [4] RunRepeat, “Brooks Hyperion Max 3 Review.”
- [5] Doctors of Running, “Brooks Hyperion Max 3 Review (2025).”
- [6] On, “Cloudsurfer Next.”
- [7] Doctors of Running, “On Cloudsurfer Next Review (2025).”
- [8] Saucony, “Endorphin Speed 5.”
- [9] Sole Review, “Saucony Endorphin Speed 5 Review.”
- [10] adidas, “Adizero Boston 13.”
- [11] RunRepeat, “Adidas Adizero Boston 13 Review.”
- [12] Nike, “Zoom Fly 6.”
- [13] RunRepeat, “Nike Zoom Fly 6 Review.”
- [14] Running Shoes Guru, “Nike Zoom Fly 6 Review.”
- [15] New Balance, “FuelCell SuperComp Trainer v3.”
- [16] Doctors of Running, “New Balance FuelCell SC Trainer v3 Review (2024).”
- [17] REI, “When to Replace Your Running Shoes.”

Appendix III: User Quotes

“At scale, every material decision becomes a financial decision, a manufacturing decision, and a sustainability decision at the same time.”-Aaron Hankins, System Innovation and Material Development Engineer, Nike

“My shoes typically last me about 400 miles, and that is 3-4 months or one cycle or maybe less of marathon training for me” – Elizabeth Fine, Marathon Runner, Cornell University

“Point and toe box strength and durability are the most important factors for training and racing shoes where a \$300+ shoe gives you about 350 miles of wear.” – Meghan Thomas, Marathon Runner, Senior Customer Experience Associate at Strava

“Running is hard on the body ... better cushioning is ideal for my safety.” - Ella Yellin, Marathon Runner (World Majors)

Appendix IV: GTM Strategy

GTM Strategy:

RegenStride's first customer is not the runner, but rather the **materials and product development engineers at a single, elite athletic footwear OEM.**

Why Nike first?

Their deep R&D infrastructure is capable of integrating novel polymer materials and have established limited-edition playbook for our validation launch strategy. Global branch reach to amplify Phase 2 mass market adoption.

Demonstrable Customer Interest

“Running is hard on the body ... better cushioning is ideal for my safety.” - Ella Yellin, Marathon Runner

- Over 50 Interviews conducted with runners, footwear professional, OEM Engineers and Footwear Product Managers
- 0 competing products offer retained cushioning as a core value proposition – open market confirmed by interviews



Total Addressable Market:

- Global Athletic Footwear Market: ~\$131B in 2024 with growing CAGR over 5%

Serviceable Addressable Market:

- Where RegenStride's extended cushioning life would thrive → Nike Shoes Market: ~\$51B (2024) based on Nike Revenue

Serviceable Obtainable Market:

- Licensing revenue capturable via single OEM partnership across 2-3 product lines in years 1-5
- Phase 1 (Yr 1-2): 1,000 limited-edition units * \$1,000 retail → ~\$80-100K in Yr 1 revenue
- Phase 2 (Yr 3-5): Integration into core OEM product lines at 100K-1M units/year * 8-10% licensing fee ~\$17.5M

Appendix V: Revenue and Breakeven

Key Metrics

Revenue Model:

- Licensing: 8-10% of wholesale price
 - ~\$5.6-\$7.00 per unit
- Single OEM → Multi product scaling

Go-to-Market Strategy:

1. Phase1: Premium limited release (validation)
2. Integration into core product lines
3. Expansion across SKUs

Unit Economics (Single OEM):

- Validation phase: 1,000 unit premium release
- Initial → actual production scale: 100k-1M units/year
- Annual revenue: \$1M-7M/year
- LTV: \$5M-25M

Customer Economics:

- \$80k-\$150k
- LTV:CAC ~ 1:7 - 1:10

Key Drivers: SKU expansion within OEM, units per product line, royalty per unit, speed of material adoption post-validation

Revenue Projections

Year	Units	Royalty/Pair (8-10%)	Revenue Range	Adoption Phase
1	1,000	\$5.60-\$7.00	\$5,600-\$7,000	Premium Halo Launch
2	100,000	\$5.60-\$7.00	\$560K-\$700K	Single Model Integration
3	350,000	\$5.60-\$7.00	\$1.96M-\$2.45M	Category Expansion
4	750,000	\$5.60-\$7.00	\$4.2M-\$5.25M	Multi-Model Expansion
5	2,500,000	\$5.60-\$7.00	\$14M-\$17.5M	Multi-SKU Platform

Material Costing

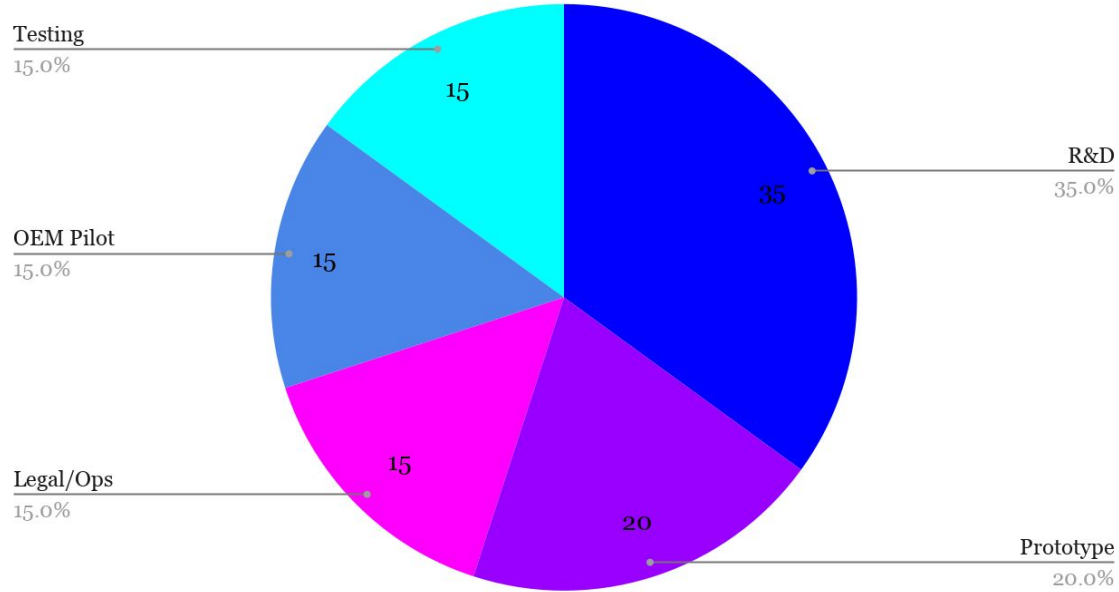
Scale	Material cost/kg	Grams/pair	Cost/pair
Y1	~\$40/kg	150g	~\$6.00
Y2	~\$15/kg	150g	~\$2.25
Y3-Y4	~\$10/kg	150g	~\$1.50
Y5	~\$6/kg	150g	~\$0.90

Appendix VI: Conferences to Attend

- **The Running Event (TRE)**: Dec 1-3, 2026, San Antonio, TX. The premier trade show for running specialty retailers and brands.
- **FFANY Market Week**: June 1–5 & August 3–7, 2026, in New York City; key footwear industry buying event
- **MAGIC Las Vegas**: Aug 10–12, 2026 in Las Vegas, features extensive athletic and casual footwear collections
- **FDRA Shoe Sourcing Summit**: July 16, 2026, NYC, focused on footwear supply chain and sourcing.
- **NW Materials Show**: September 2-3, 2026 in Portland, OR and **NE Materials Show**: August 26-27, 2026 in Danvers, MA
- **FDRA Shoe Sourcing Executive Summit**: July 16, 2026, New York City, connects footwear executives, sourcing professionals, and manufacturers to discuss supply chain strategies and partnerships.
- **SOURCING at MAGIC Las Vegas**: August 10-12, 2026, large gathering in North America featuring footwear manufacturers, suppliers, and technology providers, ideal for connecting with OEM producers.
- **Footwear Show New York Expo (FSNYE)**: June 2-4 & Dec 1-3, 2026, in NYC, B2B event, allows interaction between designers, manufacturers, and retailers

Appendix VII: Use of Funds

Use of Funds (\$750K-1M)



Appendix VIII: Patent Information

US Patent 10,745,509 B2: Methods of Making Mechanoresponsive Healing Polymers Gordon & Siochi · NASA · Filed Jul 11, 2014

What it is: PBG-BCB-PBG: A mechanophore-linked polymer that self-heals when mechanical force is applied (e.g. impact, crack)

How it works:

- Damage applies force to the polymer backbone
- The BCB (benzocyclobutene) mechanophore undergoes electrocyclic ring opening
- Covalent bonds re-form, restoring structural integrity in <100 microseconds

The patent claims 3 synthesis routes:

- Emulsion route: combine acrylonitrile + methacrylate → end-cap with bromobenzocyclobutene → block graft poly(butadiene-co-acrylonitrile)
- Anionic route: epoxidize alkene bond → ring-open with chlorosilane → living anionic polymerization → end-cap with bromocyclobutene
- Vulcanization route: free radical crosslinking with bis(triethoxysilylpropyl)disulfide at 125°C under N₂ → end-cap with bromocyclobutene

Key specs:

- Healing time: <100 microseconds
- Tensile modulus: 3.3 GPa (suitable for structural/composite applications)
- PBG composition: 75% acrylonitrile, 25% methacrylate · Block graft ratio: 13:1
- Activates at 50°C+ (vs. DuPont Surlyn® which is not load-bearing)